Silent, Plausible, and Signed: Eleven Measured Failure Modes in One Operational Marine Machine-Learning Pipeline

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Abstract—An operational machine-learning system rarely fails by crashing; it keeps running, produces output of the right shape and units, and looks correct. We report 11 instances of exactly that from one running marine data-and-modelling platform, each caught not by a failing test but by a second, cheap check most pipelines never run: a simpler baseline, a control arm, a hand-computed reference value, a synthetic fixture with a known answer, or an adversarial fixture built to break the code a specific way. None raised an exception or halted a pipeline, and every one biased its host metric in the direction that makes the system look better, or at least normal. We organise the eleven into three branches of how the failure stays hidden — no signal survives at all; a real, correctly computed number is licensed by convention to mean more than it does; or the behaviour is textbook and expected, so its downstream cost is never re-examined — and map each instance to the specific check that caught it. Because a paper about untrusted plausible numbers cannot itself rest on unverified prose, we grade every instance's own evidence (a committed artifact versus a dated, reproducible-on-demand measurement) and report two of our own numbers caught drifting while writing this paper, including a transcription error in a companion manuscript, as further, unplanned instances of the same thesis.

Index Terms—software engineering for machine learning, data quality, silent failure, technical debt, reproducibility, marine data systems

I. Introduction

An operational machine-learning system rarely fails by crashing. It keeps running, produces output of the right shape and the right units, and looks correct — and the failure that matters is the one that never trips an exception, an alert, or a test that was already passing. This paper reports eleven such failures, each one found and fixed inside a single running platform (marine ocean-data ingestion, forecasting, and species-distribution and harmful-algal-bloom modelling), and each one caught not by monitoring, not by a test that failed, but by a second, cheap check that most pipelines never run: a comparison against a simpler baseline, a control arm, a value computed by hand from raw data, a synthetic case with a known correct answer, or a fixture built deliberately to break the code in a specific way. None of the eleven raised an exception or halted a pipeline. Every one biased its host metric in the direction that makes the system look better, or at least look normal, than it is.

This is not a paper about one bug. It is a paper about a shape that recurs: a system component that is individually well-tested and behaves exactly as documented, feeding a downstream reading that is technically a real number computed from real data, which is nonetheless the wrong number to trust for the question being asked of it. Sculley et al. [1] named the general phenomenon at the level of system architecture — machine learning systems accrue a form of technical debt largely invisible to the metrics that are usually watched. Sambasivan et al. [2] documented it at the level of organisational practice — data problems compound silently across a pipeline because "data work" is systematically undervalued relative to "model work." This paper sits between those two: eleven concrete, measured instances from one running system, each with the specific check that caught it named and, where the evidence supports it, reproducible on demand.

A. Contributions

1) Eleven measured instances of silent, plausible, signed-direction failure, drawn from production code rather than constructed as illustrations, spanning data ingestion, geometric resampling, ensemble modelling, detector design, and cross-region evaluation.
2) A three-branch taxonomy of how the failure stays hidden: no signal survives at all (§IV-A); a real, correctly computed number is licensed by convention to mean more than it does (§IV-B); or the behaviour is documented and expected, so its downstream consequence is never re-examined (§IV-C).
3) An explicit mapping from each instance to the check that caught it (Table I), because "run a second, cheap check" is only actionable advice once the check itself is named.
4) A reproducibility ledger, applied to this paper's own claims: six of the eleven instances are fully backed by a committed report or a currently-passing regression test; three combine a tested mechanism with a headline number that is a dated, single live measurement

of a state that moves (a current field, a wind field, a just-retrained model); and two rest entirely on a dated measurement with no committed artifact at all, each naming the script that reproduces it and reported as of its date rather than as a standing fact (§III). Two of those dated numbers were caught drifting while writing this paper, which §V reports as a further instance of the paper's own thesis rather than an inconvenience to edit around.

II. Related Work

a) Technical debt in ML systems: Sculley et al. [1] catalogue system-level hazards — boundary erosion, hidden feedback loops, undeclared consumers, glue code — that accumulate in production machine learning without tripping any single component's tests. That paper argues at the level of system design; this one operates one level down, at the level of a specific number a specific function returns, and asks what second check would have shown that number to be misleading. Several of the instances here are concrete specialisations of debt classes that paper names only abstractly: I.9's coverage field and I.4's aggregation ordering are both "hidden feedback" in the sense that the input's own geometry silently determines what a downstream consumer receives.

b) Data cascades: Sambasivan et al. [2] define a data cascade as a compounding, downstream effect of a data-quality issue that is invisible at its point of origin and surfaces far away, in high-stakes deployment contexts. I.4 (an instantaneous sample standing in for a period mean), 2.5 (a memory-motivated batching change that is not value-identical to the path it replaces), and I.9 (a resampling routine that erases the coastal band it was asked to preserve) are data cascades in exactly that sense, with one addition: each is quantified with a specific, measured cost — a skill delta, an error magnitude, a cell count — rather than reported as a category of risk. Their interview-based methodology surfaces that data work is undervalued across many teams; this paper's contribution is complementary evidence of what that undervaluing costs, itemised, in one system where it was possible to measure.

c) Scope: This is a single-system instance catalogue, not a general taxonomy validated across projects, and §VI says so directly. Two of the eleven instances are the headline results of dedicated companion papers from the same codebase — ensemble weighting (I.1) and the global-versus-regional habitat comparison (I.2, [3]) — and are summarised here rather than re-derived; a third companion paper [4] establishes the persistence-baseline finding this paper's framing extends to eleven further instances outside forecasting alone.

III. Method: how these were found, and how each is graded

a) The five checks: Across the eleven instances, the check that surfaced the problem falls into one of five kinds, and more than one recurs per instance: (1) a second baseline or reference — scoring an ensemble against its own best member, or a filtered series against its unfiltered one; (2) a control arm — measuring a signal where it is not expected to hold, alongside where it is; (3) a hand-computed value against raw data — counting erased cells directly, or computing a per-product dispersion by hand; (4) a synthetic fixture with a known correct answer — a diurnal cycle whose true mean is known by construction, a batching path checked against its own unbatched reference; and (5) a fixture built deliberately to break the code in a specific, adversarial way — a vortex placed on the antimeridian, a coastline placed to the west instead of the east, a field of pure white noise. None of the five is expensive. All five are absent by default from a pipeline whose own tests pass and whose own output looks plausible.

b) Evidence grading: A paper whose thesis is "the plausible number is not the one to trust" cannot itself rest on unverified prose, and this project's own written history already contains a caution that applies directly here: a dated note recorded during a previous session is a claim about that session, not a standing fact, and has been wrong before after re-investigation. Accordingly, every instance below is marked as backed either by a committed artifact — a report file or a currently passing regression test that reproduces the number on demand, re-verified as part of writing this paper on 2026-09-09 — or by a dated measurement: a single, live run recorded with its date, naming the script that would reproduce it, with no output file committed. Six instances (I.2, I.4, I.5, I.9, I.13, 2.5) are fully artifact-backed — every sub-claim traces to a report file or a currently-passing test, with nothing resting on a value that could differ if re-measured today. Three (I.1, I.8, I.11) pair a tested mechanism with a headline number that is exactly that kind of value: a live measurement of something that keeps moving — a just-retrained model's cross-validated score, a day's ocean-current field, a day's wind field — rather than a fixed model artifact. The remaining two (I.7, I.12) rest entirely on a dated measurement, with no committed test covering even the mechanism; two of the dated numbers across these two groups (I.1, I.12) turned out to matter, as §V reports.

c) System: All eleven instances come from MarisAI, an ocean-data platform combining live provider ingestion (satellite and reanalysis products), a forecasting engine trained per-variable per-site, a detector layer (eddies, upwelling, marine heatwaves), and offline species-distribution and harmful-algal-bloom pipelines sharing a common feature-store spine. No architectural detail beyond what a given instance needs is given here; the companion papers [3], [4] describe the forecasting and habitat-modelling components respectively in full.

IV. Results: eleven instances

A. Branch A: no signal survives at all

In these three, the wrong answer produces no exception and, for most of the symptoms described, nothing a bounds check would catch either — it is a real number in the right units, silently computed from the wrong quantity. The one exception inside this branch is named where it occurs (I.5's colour- scale symptom): even there, nothing stopped the pipeline, because nothing checks a percentile against a circular domain.

a) I.4 — a period mean silently becomes an instantaneous sample.: Joining an hourly product against a daily one and aggregating afterwards substitutes the join's arbitrary matched timestamp for a period mean, with no signal that a substitution occurred. On a synthetic 3.0°C-amplitude diurnal cycle around a 20.0°C mean, the 00:00 sample is exactly 23.0°C — the full amplitude away from the true daily mean, in a signed direction fixed by what time of day the join happens to land on. A companion defect in the same code path is non-linear: a current rotating at a constant 1 m/s speed averages its u/v components to ≈ 0 before the speed is derived, reporting slack water for a current that never stopped moving, unless speed is derived before the aggregation rather than after (verified: a full daily rotation at unit speed derives to 1.0 m/s only when ordered this way). The fix reorders resolution, aggregation, and merge so the coarse product's own cadence determines the join, and a regression test builds exactly this scenario.

b) I.5 — a circular variable breaks at every layer that treats it as linear.: A bearing on $[0, 360)$ has the property that 359° and 1° are two degrees apart, not 180° apart — and every stage of a pipeline that is not built for that property gets it wrong the same way. Naive linear interpolation of those two headings returns 180°, the exact reverse of both inputs; this is fixed by resampling in sin/cos space and recombining with atan2. A forecast difference needs the analogous fix, a signed veer wrapped to $[-180, 180)$, or a percentile-based colour scale computed on a circular domain silently ships a range like display_max: 400 on a field whose true maximum is 360°. A fourth layer sits underneath the first three and is the one that is easy to miss entirely: a forecasting model trained with target_mode: delta learns the numeric change in the bearing, and a 5-degree veer across north trains as -355 — teaching the model an enormous excursion that never physically happened, while the model still fits, still scores, and still produces bearings in range. The fix for this fourth layer is not a wrapping correction but an architectural one: forecast the vector components and derive the bearing, exactly as the codebase's own particle-advection layer already did for unrelated reasons, which is precisely why that layer was never affected while the raster direction layer was. This closes the gap for wind and current direction, both of which have components to decompose into; wave direction has no such components in this system and remains linearly trained, a disclosed, deliberate asymmetry rather than the same open gap under a different name.

c) 2.5 — a memory-saving batch is not automatically value-identical.: A global point-sampling job batched by calendar year, built because the unbatched path exhausts memory on a resource-constrained machine, looks like a pure performance change: same inputs, same code path per batch, smaller peak memory. It is not value-identical unless two specific details are handled. First, source products are stamped mid-month with nearest-in-time selection, so a point near a year boundary can have its true nearest timestep fall in the adjacent year; without a temporal pad, a batch boundary silently substitutes the wrong timestep. Second, one derived quantity (eddy kinetic energy) is computed against the whole record's mean, which a per-year batch would instead compute against that year's own mean under an identical column name, corrupting the value while leaving its label unchanged. With the padding at 45 days, the equivalence holds to within the batching path's own test tolerance (10^{-6}, relative and absolute); reproduced directly for this paper by removing the pad and re-running that same equivalence fixture, the maximum absolute error on one variable (thetao) is 2.92°C, affecting precisely the two rows that sit at a year boundary — silent because a batching refactor is not the kind of change anyone expects to alter a value at all.

B. Branch B: a real number, licensed to mean more than it does

In these five, the number itself is computed correctly. What is silent is the unlicensed step between the number and the conclusion drawn from it.

a) I.1 — an ensemble's own accuracy score does not certify that combining helped.: Weighting an ensemble's members proportionally to a skill score whose chance level is zero compresses a large quality gap into a small weight gap. Measured 2026-08-13: MaxEnt's cross-validated true skill statistic (TSS) of 0.619 against LightGBM's 0.826 — less than half as good on the holdout — still drew 27% of the proportional vote, and the resulting ensemble scored 0.694 holdout TSS, below its own best single member's 0.788. Softmax weighting over the same cross-validated scores raises the ensemble to 0.792, above every member, at a stated cost in spatial calibration (Boyce index 0.936 $\rightarrow$ 0.905). The check that catches this is not a property of the ensemble at all — it is scoring the best single member as its own baseline, which most reported ensemble results never do.

b) I.2 — more training data does not certify improved skill on the water that matters.: A fish-habitat model's own cross-validated score improves when its training extent grows from one regional sea to the whole global ocean, using 64.7× more rows (1,902 to 123,104). On a matched holdout — identical regional rows, scored by both

models, with the holdout's own spatial blocks removed from the global model's training set — the ranking reverses: regional TSS 0.798 against global 0.448, and a Multivariate Environmental Similarity Surface analysis rules out extrapolation as the explanation (the global model's holdout rows sit more centrally in its training envelope, 1.24% outside versus the regional model's own 9.04%). Full mechanism diagnosis (a SHAP-based feature-displacement account, confirmed by ablation) is the subject of [3]; the instance is included here because it is the same shape as the other ten: a real, correctly computed number — the global model's own cross-validated score — licensed by convention ("more data helps") to mean something a matched comparison shows it does not.

c) I.8 — a detection count is not a census.: An Okubo–Weiss eddy detector applied to the live global current field returns 2,097 features (954 cyclonic, 1,143 anticyclonic; median radius 55 km), densest exactly where oceanography predicts (Gulf Stream, Kuroshio, Agulhas, the Somali Great Whirl). Every one of those detections is a real, reproducible feature of a variance-relative threshold applied to a specific field snapshot — and that is precisely what a count is not licensed to be read as: a fixed inventory of how many eddies exist. Run the identical detector on a field of pure white noise and it still returns features, every one of them pinned to the algorithm's radius floor — a large detection is evidence of real structure, a small one may not be, and only the adversarial fixture (noise with a known-zero true answer) distinguishes the two. Two further defects are the kind that would silently corrupt a genuine detection rather than merely mislabel a count: polarity must be read as $\mathrm{sign}(\zeta) = \mathrm{sign}(\text{latitude})$, not the vorticity sign alone, or a detector is confidently wrong in one hemisphere; and the antimeridian breaks derivative computation, connected- component labelling, and centroid arithmetic independently, the last of which averages $+179°$ and $-179°$ to $0°$ and relocates a Pacific eddy to the Gulf of Guinea unless wrap-aware arithmetic is used throughout. All three defects are covered by regression tests; the headline count is a snapshot of a live current field, not a fixed artifact, and is reported here as such.

d) I.11 — a corroboration rate is not evidence of causation without a control arm.: An upwelling index derived from wind stress is corroborated against an independent cold-SST-anomaly signal: on the live global field, measured 2026-08-17, 19.9% of upwelling-favourable coastal cells were anomalously cool for the season. Read alone, that number reads as confirmation. Scored identically on downwelling-favourable coasts, where cool water is not expected at all, the control arm returns 17.2% — and below the seasonal 10th percentile the two are statistically indistinguishable (4.1% against 3.9%), with favourable coasts on average warmer ($+0.91°$C) than downwelling ones ($+0.60°$C). The corroboration is a real observation of cool water and a weak coincidence, not evidence the wind drove it, and the platform now computes and returns the control fraction with every response rather than leaving it for a reader to think to ask for. A related regression test is itself an instance of the same discipline applied at the unit level: a coastline fixture placed to the west of open water, called an eastern boundary, catches the one sign error that would have silently reported every upwelling coast on Earth as downwelling.

e) I.13 — a precision number does not rank model skill without controlling for base rate.: Harmful-algal-bloom early-warning precision at a fixed recall is a standard way operational systems and papers alike report performance, region by region. Across five independently trained regions, precision at recall 0.8 orders almost exactly by each region's own bloom base rate. The Arabian Sea (base rate 0.076) and the Bay of Bengal (0.078) sit at nearly identical prevalence and report nearly identical t+7 precision (0.202 against 0.187) — and yet the Bay of Bengal carries roughly $0.150/0.053 \approx 3\times$ the skill lift over a persistence baseline at the same horizon. Same apparent performance, measurably different models: precision alone cannot tell them apart, and the check that does is a second baseline (persistence) reported beside it rather than instead of it.

C. Branch C: visible, expected, and therefore never re-examined

In these three, the behaviour that causes the problem is not hidden — it is textbook, or it is an obviously reasonable-sounding simplification. What is silent is that its downstream consequence was never checked.

a) I.9 — documented NaN behaviour, undocumented coastal erosion.: Bilinear interpolation returning NaN when any of its four surrounding cells is off the ocean mask is exactly the textbook behaviour of bilinear interpolation over a masked field; nothing about it is a bug in isolation. Applied to render a forecast grid onto tile pixels, the consequence is that the painted ocean retreats by up to a full grid cell from every coastline, producing roughly 110 km of axis-aligned staircase at $1°$ resolution and erasing 3,609 of chlorophyll's 42,499 ocean cells (8.5%) — exactly the coastal band the layer exists to show. Nobody who understands bilinear interpolation is surprised that a masked neighbour poisons the result; the gap is that the erosion was never counted until it was counted directly, cell by cell, against the raw grid. The fix separates a nearest-filled values field (so coastal cells stay finite) from a 0/1 coverage field interpolated the same way and thresholded at 0.5, so the rendered edge follows the bilinear half-contour rather than the mask's own axis-aligned steps.

b) I.12 — a baseline fitted on one product is assumed to transfer to another.: Scoring a fresh observation against a long-record climatology fitted on a different data product is treated, by convention, as a substitution that only affects timeliness — the fresher product is newer and higher-

resolution, so replacing a 14.5-day-old satellite record with an hourly reanalysis field reads as a strict improvement. Measured 2026-08-17 on an identical wind field, contrast defined as favourable-minus-control: the weak, mean-based corroboration tier was essentially unchanged (0.026 against 0.022), while the strong, percentile-based tier inverted, from 0.002 to -0.149. The cause, measured separately by comparing one day of both products on the same grid, is not a systematic offset (mean 0.033°C in open water) but per-cell dispersion — standard deviation 0.467°C in open water, rising to 0.758°C across the coastal band, where 24.9% of cells differ by more than 0.5°C and 10.0% by more than a full degree — wider than the 0.5°C threshold the tail test is built around. A mean-based test degrades gracefully under that noise and hides the problem; a tail-based test cannot survive noise the width of its own signal, and inverts instead. Two further attempts at the same fix, tried after this result, both failed the same way rather than a new way: fitting a matched climatology on the fresher product's own long record still inverted the tail test (-0.051 against the OISST arm's -0.001), and averaging the wind field over a short rolling window moved both contrasts toward zero without correcting either. The product mismatch, not the staleness it was meant to fix, is the binding constraint, and the visible-but-expected assumption it hides is that "fresher and higher-resolution" implies "safe to substitute."

c) I.7 — a standard filter is trusted precisely because it is standard.: A Hampel filter — median-and-MAD outlier rejection over a rolling window — is a conventional, textbook choice for cleaning an observational time series, and its use here is not itself unusual. Applied to a clean, already-quality-controlled Copernicus sea-surface-temperature series, it flags 5–12% of values as outliers at every window and threshold tried, because a smooth geophysical field makes the local MAD collapse toward zero by construction — of 367 daily values, 74 have a local MAD of exactly zero.[1] Disabling the filter measurably improved seven-day forecast skill, from 0.138 to 0.202. The filter is not malfunctioning — it is doing exactly what a Hampel filter does, on a field where that behaviour is a liability rather than a safeguard, and the "it's a standard technique" reasoning that makes it easy to add is the same reasoning that makes nobody check whether it is a standard technique for this kind of field.

V. Discussion

Read across all eleven, two patterns hold that neither instance alone would establish. First, the direction is never neutral: every failure in this set biased its host metric toward looking better, or at least more normal, than the checked number shows it to be. A period mean read high rather than scattering randomly around the true value; an ensemble's headline number looked like an improvement; a detection count read as more complete than a threshold-consistent detector can support; a fresher data product looked like a strict upgrade. A failure that made a system look randomly wrong would be caught by the first user who compared two runs. A failure that makes it look plausible, or better, is caught only by whichever of the five checks in §III someone happens to run — and Table I's evidence column shows that even inside one team's own codebase, most of those checks were run once, dated, and not committed as a standing artifact.

Second, that observation applies to this paper's own evidence, not only to the systems it describes, and two instances demonstrate it directly rather than by analogy. Verifying I.2 for this paper meant re-reading the matched-holdout comparison's own source file rather than trusting the companion manuscript's macro, and the two disagreed: the companion manuscript states the regional MaxEnt tier's TSS as 0.407, and the source file gives 0.365 — a transcription slip from spreadsheet to LaTeX macro that nothing had re-checked since the macro was written, because 0.407 is exactly as plausible as 0.365 and neither is used elsewhere in that manuscript's argument. It has been corrected in that manuscript's own build as a direct consequence of applying, to this paper's own citations, the same discipline §III asks of the systems being studied. I.1 makes the same point from a different angle: the 2026-08-13 measurement quoted in §IV-B is not a standing fact about the current model. Re-read on 2026-09-09, MaxEnt's cross-validated TSS has since risen to 0.689 (from 0.619), narrowing the quality gap the whole failure mode depends on, and the current ensemble weights it accordingly (50.2%/46.5%/3.3% lightgbm/random forest/maxent). The mechanism the original measurement demonstrated — proportional weighting can score an ensemble below its own best member — still holds on an unrelated, more recent holdout, at a much narrower margin (ensemble TSS 0.8554 against the single best member's 0.8564), which is itself evidence the effect's size is a direct function of the between-member quality gap it was always claimed to depend on. Neither correction changes which of the eleven instances is real; both show that a catalogue of silent, plausible, signed-direction failures has to keep re-deriving its own numbers rather than treating a prior write-up, including its own companion papers, as a settled source.

a) A practical checklist: The five checks in §III are not five different disciplines requiring five different tools — they are five answers to one question, asked of any number a pipeline is about to trust: what would this number look like if it were wrong in the specific way this computation could be wrong, and is that comparison actually run. A

[1]The documentation trail for this instance is internally inconsistent on this one number: the surrounding prose in this project's own notes rounds the window to 365 days, while the filter's own code comment gives 367. 367 is used here as the figure closer to the implementation that produced it, and the inconsistency is left visible rather than silently resolved, since a paper about undetected numeric drift should not quietly paper over an instance of exactly that in its own source material.

TABLE I
All eleven instances: what the plausible reading obscures, the check that surfaced it, and this paper's own reproducibility ledger for the claim (Artifact = every sub-claim traces to a committed report or a currently-passing test; Artifact + dated = a tested mechanism paired with a headline number that is a single, named, re-runnable measurement of a moving state, with no stored output; Dated = no committed artifact at all, mechanism included).

#	Reads as	What the check found	Evidence
I.4	a period mean	an instantaneous sample, off by the full diurnal amplitude	Artifact
I.5	a bearing handled correctly	linear treatment reverses, wraps, or mis-scales it at up to four layers	Artifact
2.5	a pure performance optimisation	a non-equivalent result at year boundaries	Artifact
I.1	an ensemble that improved on its members	an ensemble that scored below its own best member	Artifact + dated (2026-08-13)
I.2	more data, therefore more evidence	less skill on the region the model serves, despite 64.7× the rows	Artifact
I.8	a census of ocean eddies	a threshold-consistent detector, not a count; returns features from noise	Artifact + dated (count)
I.11	confirmation that wind drove an SST anomaly	a rate barely above its own control arm	Artifact + dated (2026-08-17)
I.13	model skill, comparable across regions	regional base rate, not skill, ordering the metric	Artifact
I.9	full ocean coverage to the coastline	8.5% of coastal cells erased by a textbook NaN rule	Artifact
I.12	a strict improvement (fresher, sharper product)	an inverted tail test from per-cell product disagreement	Dated (3 attempts)
I.7	a safe, standard cleaning step	5–12% of clean values flagged; skill cost measured	Dated

second baseline answers it for "did the combination help." A control arm answers it for "does the corroborating signal mean what it is being read to mean." A hand-computed value against raw data answers it for "does the pipeline's aggregate match what direct inspection shows." A synthetic fixture with a known answer answers it for "is this transformation actually equivalent to the one it replaced." An adversarial fixture answers it for "what does this component do on an input built to break exactly this."

VI. Limitations

This is an instance catalogue from one system, assembled by the team that built it, not a survey validated against an independent sample of ML pipelines; the three-branch taxonomy in §IV is inductive from these eleven and has not been tested against failures found elsewhere. Most of the eleven were found while building or extending a feature, not while auditing for bugs specifically, which is itself informative about base rate but is not a controlled search and cannot support a claim about how common this shape of failure is across the field. Five of the eleven instances rest at least partly on a dated, single measurement rather than a committed artifact: three (I.1, I.8's headline count, I.11) pair a headline number that is a dated measurement of a moving state with a mechanism that does have its own passing regression test, and two (I.7, I.12) are dated throughout, with no committed test covering even the mechanism. §III and Table I say which is which per instance rather than averaging that uncertainty away, and a reader should weight the specific numbers in those five accordingly. I.1 and I.2's fuller mechanism is documented in companion papers cited rather than reproduced in full here; this paper's contribution for those two is the cross-cutting frame, not new evidence beyond what §V adds.

VII. Conclusion

Eleven instances from one operational marine machine-learning platform share a shape: a component that is individually correct, well-tested, and documented, feeding a downstream number that is real, computed from real data, and licensed by convention to mean more than a second, cheap check shows it to mean. None produced an exception; all biased their host metric in the flattering direction. The checks that caught them — a second baseline, a control arm, a hand-computed reference, a synthetic case with a known answer, an adversarial fixture — cost little and are almost never run by default. Two of this paper's own numbers drifted between when they were first measured and when this paper verified them, which is not a flaw in the argument but its clearest illustration: the discipline this paper recommends for reading a pipeline's output has to be applied to a paper's own citations, continuously, not once.

Reproducibility

Every macro in shared/generated/macros.tex names its source file and evidence grade inline; the figure is generated by generate_figures.py in this paper's shared/ directory. Every number below was verified against the file naming its source as of 2026-09-09.

Artifact (fully backed). I.4, 2.5: backend/tests/test_download_cadence.py, machine_learning/tests/test_global_sources.py. I.5: backend/tests/test_field_sampling.py, backend/tests/test_derived.py, backend/services/download/registry.py. I.9: backend/services/

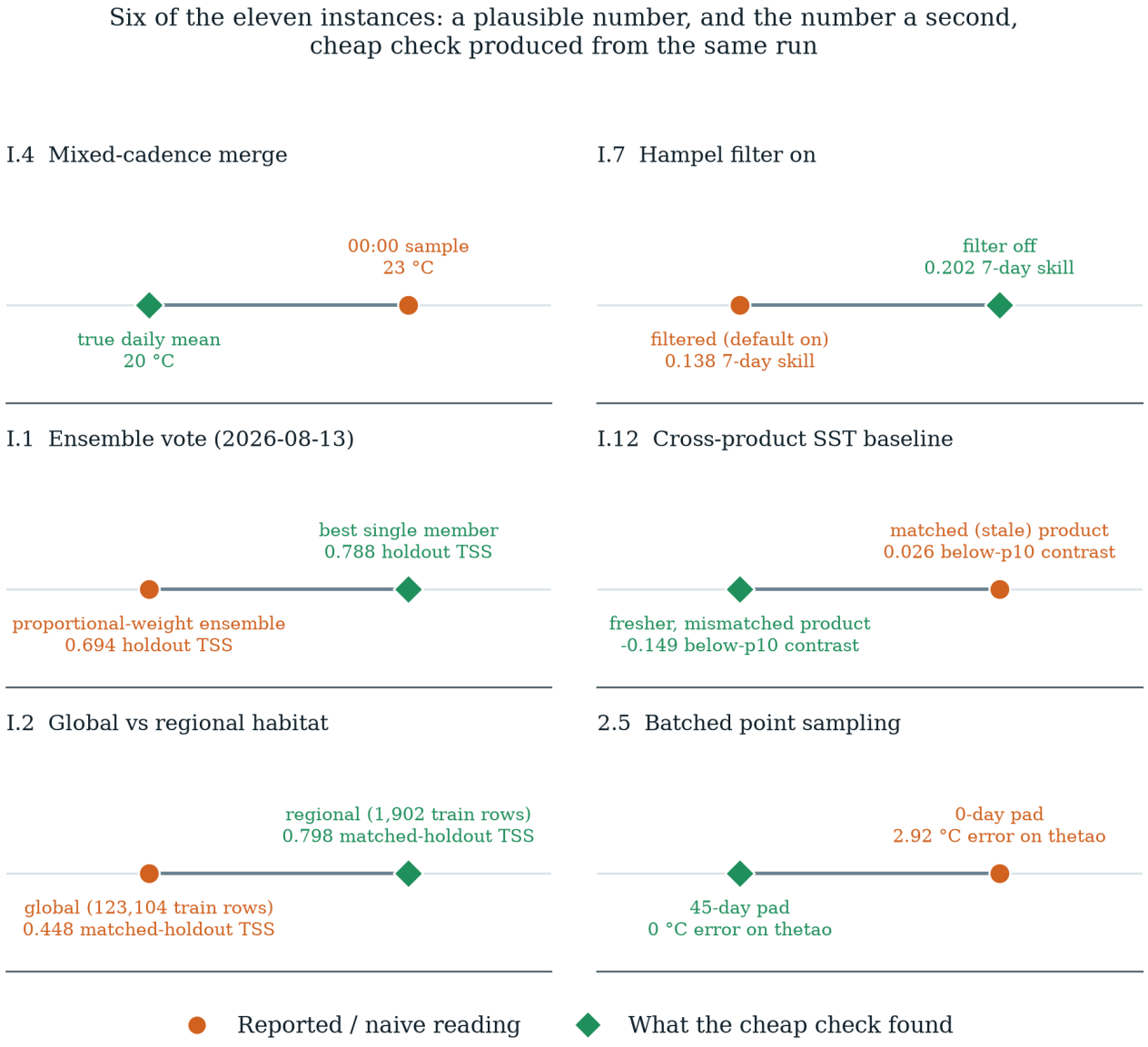


Fig. 1. Six of the eleven instances reduced to a single before/after number pair: the reading that looked plausible on its own (orange circle) against what a second, cheap check produced from the same run (green diamond). Units differ by row; each panel has its own scale.

field_sampling.py, backend/tests/test_forecast_grid.py. I.13: machine_learning/reports/hab_early_warning_*_summary.json and matching *_holdout.csv for all five regions. I.2 (re-verified 2026-09-09): machine_learning/reports/global_vs_regional_habitat.csv; I.2's full mechanism is machine_learning/scripts/validate_global_on_region.py [3].

Artifact + dated (tested mechanism, dated headline). I.1: mechanism and current re-verification (2026-09-09) in machine_learning/reports/fish_habitat_summary.json, fish_habitat_holdout.csv; 2026-08-13 headline figures in machine_learning/fish_habitat_prediction/src/models.py. I.8: mechanism tests in backend/tests/test_eddies.py; dated count in backend/services/eddies.py. I.11: mechanism test in backend/tests/test_upwelling.py; dated corroboration rates in backend/services/upwelling.py.

Dated (no committed artifact). I.7: backend/forecasting/preprocessing.py. I.12: backend/services/sst_anomaly.py, naming scripts/measure_sst_corroboration.py and scripts/build_climatology_copernicus.py.

References

[1] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Den-

nison, “Hidden technical debt in machine learning systems,” in Advances in Neural Information Processing Systems 28 (NIPS 2015), 2015, pp. 2503–2511.

[2] N. Sambasivan, S. Kapania, H. Highfill, D. Akrong, P. Paritosh, and L. M. Aroyo, “”everyone wants to do the model work, not the data work”: Data cascades in high-stakes ai,” in Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI ’21), 2021.

[3] D. Krishna, “More data, worse water: When expanding a species distribution model’s training extent degrades skill on the region it serves,” 2026, manuscript, MarisAI.

[4] ——, “Rising skill, falling skill: Why persistence alone is an insufficient baseline for statistical ocean forecasting,” 2026, manuscript, MarisAI.